In order to achieve the specified thickness, a typical specification may require that spread rates fall between the target and 10 percent over the target.

10.6.5 Pavement Smoothness

Pavement smoothness for quality-control purposes is generally measured with a straightedge. Typical acceptance specifications require hand measurements with a 10-foot (3 meter) straightedge, a rolling straightedge (also called the California profilograph) or, inertial profilometers which measure the International Roughness Index (IRI).

Typical requirements with a 10-foot straightedge are that there be no bumps or dips greater than ⅛ inch when compared with the two outer edges of the straightedge. Measurements are generally taken in both the longitudinal and the transverse direction. The straightedge is laid on the pavement surface, and the maximum deviation of the surface from this straightedge is measured and recorded. These measurements can be taken while compaction is ongoing, and feedback can be provided to the paver and the roller operators for making adjustments.

The rolling straightedge, or profilograph, is used by a large number of states to measure the initial roughness of new construction and overlay projects. These profilographs use wheels at each end of the straightedge as the contact point on pavement surface, and deviations in the pavement profile are measured at the center point between the wheels. The devices can generally be used in lengths of 10–30 feet (3–9 meters). The measured deviations are accumulated and reported as roughness in inches per mile. Profilograph measurements and readings can be taken while compaction is ongoing or immediately afterward. Because profilograph testing can provide quick feedback to the paving crews, such testing can be useful as a quality-control test. Some contractors will also use profilograph testing to evaluate the smoothness of a pavement before placement of the surface course. The cost of profilograph testing and subsequent grinding to improve smoothness could be inexpensive when compared with potential incentive payments.

10.7 Acceptance Point and Test Results

Acceptance testing is usually provided for the following properties in HMA project specifications:

- Asphalt content
- Aggregate gradation
- Compaction
- Volumetric properties (air voids, VMA, VFA)
- Aggregate characteristics (CAA, FAA, F&E, SE, etc.)
- Ride quality
- Pavement thickness or spread rate

The criterion for acceptance is generally the same as the parameters tested for quality-control. This makes sense because the purpose of quality-control is to control production so the final product meets the acceptance criteria. Because of differences in the use of quality-control and acceptance tests, however, there are different sampling points and differences in test procedures in some cases. The following sections discuss the sampling and testing methods commonly used for acceptance, giving emphasis to those that may be different from the quality-control methods discussed in Section 10.3.

10.7.1 Asphalt Content and Gradation

The point of acceptance for asphalt content and aggregate gradation is normally derived from the finished HMA product. HMA samples are typically obtained from the roadway or the delivery truck. Refer to Sections 10.2.3 and 10.2.4 for more detailed descriptions of these sampling methods.

Asphalt binder content may be determined by solvent extraction, nuclear, or ignition oven testing. The use of the ignition binder content method is increasingly becoming the method of choice for acceptance testing. This test provides reasonably accurate test results if it is properly calibrated for the specific mix design materials being used, and it allows the use of the recovered aggregate materials to be tested for gradation.

10.7.2 Aggregate Angularity (Fractured Faces)

The point of acceptance for aggregate angularity is normally in the roadway or after blending of the aggregate stockpiles. HMA samples from the

roadway are used for angularity testing by some agencies and must be tested after extraction and gradation. Concerns over possible effects of mixing and heating in the plant and possible residue left after asphalt solvent extractions or ignition burns have led other agencies to use the aggregate cold feed or stockpiles as the point of acceptance for aggregate angularity. Refer to Sections 10.2.3 and 10.2.4 for more detailed descriptions of these sampling methods.

Aggregate angularity is often used as an acceptance criteria for both the coarse and fine portions of the aggregate sample. The CAA test can be used for the coarse portion of the blended aggregate samples. The FAA test is often used for the fine portion of the blended aggregates. The FAA uncompacted voids test is gaining in popularity because it is simple to run, repeatable, and less prone to the judgment of the testing technician, thereby reducing the number of disputes over the test results.

The methods used for acceptance of aggregate angularity are the same as those used in quality-control testing.

10.7.3 Volumetric Properties (Voids, VMA, VFA)

The sampling point for acceptance of volumetric properties is normally at the point of delivery on-site or in delivery trucks at the plant. HMA samples are generally taken from the roadway behind the paving machine, from plant sampling devices, or from the delivery trucks prior to leaving the plant. Refer to Sections 10.2.3 and 10.2.4 for more detailed descriptions of theses sampling methods.

Volumetric properties generally refer to one or more of the following three test parameters: 1) Laboratory air voids; 2) voids in the mineral aggregate (VMA); or 3) voids filled with asphalt (VFA). These properties are calculated using values derived from several different tests. Laboratory air voids require bulk specific gravity results from the laboratory compacted specimen and the maximum theoretical specific gravity. Calculations of VMA require the following data: laboratory compacted bulk specific gravity of the mix; the bulk specific gravity of the blended aggregate; and the asphalt content as a percentage of total mix. Calculations for VFA use the air voids and the VMA values.

10.7.4 Compaction

The acceptance point for compaction is the finished pavement. Sampling is usually accomplished by cutting cores with a wet cut high-speed core drill and diamond-studded core barrels. These methods remove the cored specimens with little distortion or damage. The cored test specimens are easily damaged, and the damage is not always visible. They need

to be treated with care and kept cool to minimize chances for deformation. In the laboratory, they should be placed on a smooth table, with the flat portion of the core down, and in a cool environment until testing.

In some cases, compaction is accepted based on nuclear density results. Nuclear densities should be required to be correlated to core density tests if they are used for acceptance.

There are two ways to calculate compaction values, and they can give very different results. Compaction can be specified by the percentage of either the maximum theoretical density or the bulk laboratory density. The method specified should be clarified before bidding. Differences in compactive effort required by the methods and relative values of compaction specifications are discussed in more detail in Section 10.8.3 under "Reference Density Specifications."

Compaction Using Maximum Theoretical Specific Gravity

Compaction is usually calculated by dividing the field measurements of density (specific gravity) by the maximum theoretical specific gravity of the mix.

$$\text{Compaction} = \left[\frac{(\text{field bulk specific gravity})}{(\text{maximum theoretical specific gravity})}\right] \times 100$$

As an example, consider a project that specifies compaction of at least 93.0 percent of maximum theoretical specific gravity and the maximum theoretical specific gravity measured from a given day's production is 2.404. This specific gravity value can be converted to density by multiplying it by the unit weight of water (62.4 lb/ft^3 (1000 kg/m^3)), which yields 150.0 lb/ft^3 (2404 kg/m^3). If the acceptance testing yields an in-place density of 139.2 lb/ft^3 (2228 kg/m^3) at a given location, then compaction is calculated as follows:

$$\text{Compaction} = \left(\frac{139.2}{150.0}\right) \times 100 \text{ or } \left[\left(\frac{2228}{2404}\right) \times 100\right] = 92.8 \text{ percent}$$

This compaction test result does not meet the minimum requirements, and the specifications should be checked to see what action is required for a failing density test.

Specifications that require compaction as a percentage of maximum theoretical specific gravity typically set minimum values that range from 92 percent to 94 percent compaction. Maximum values for compaction specifications are generally set at 97 percent.

Compaction Using Bulk Laboratory Density

Compaction can also be calculated by dividing the field measurements of density (bulk specific gravity) by the laboratory bulk density (bulk specific gravity). The laboratory density will be measured by Gyratory, Marshall, or Hveem methods after the specimens are compacted. The calculation is simply:

$$\text{Compaction} = \left[\frac{(\text{field bulk specific gravity})}{(\text{lab bulk specific gravity})}\right] \times 100$$

Specifications that are written as a percentage of bulk density typically use minimums that range from 95 percent to 98 percent compaction.

10.7.5 Thickness

The acceptance point for thickness is the finished pavement. It is almost exclusively measured by cutting cores from the completed pavement surface. The cored specimens taken for the compaction testing described above are measured for thickness before testing for unit weight.

Some agencies specify spread rates in lieu of thickness. Spread rates can be checked as either an acceptance criteria or as a quality-control measure by calculating the weight of material placed over a measured area. For example, if the paver is placing a 14-foot (4.3-meter) wide mat and it spreads a 22.0-ton load over a length of 100 feet (30.5 meters), the spread rate would be 31.4 pounds per square foot. Assume the target-spread rate was 30.2 lb/ft^2 (147.5 kg/m^2) to achieve an average design thickness of 2½ inches (63.5 millimeters). The actual spread rate in this example is 4 percent over the target-spread rate.

Agencies do not like to have thinner pavement sections than are specified in the design, so a typical specification would require that spread rates fall between the target and 10 percent over the target.

Spread-rate specifications are especially useful for overlay projects that are seeking smoothness improvements on the roadway. Paving equipment, by its nature, provides significant leveling characteristics as it places the asphalt thicker in low areas and thinner in high areas. This improves ride quality but creates variations in pavement thickness. The more rough the existing pavement, the more variation in pavement thickness.

10.7.6 Ride Quality

Pavement smoothness is the leading factor in the traveling public's perception of road quality. In addition, rough or uneven pavements can have an adverse affect on fuel usage, roadway safety, and vehicle maintenance costs. Smoother pavements can also improve pavement durability, thereby extending the life of the pavement.

Many agencies use pavement smoothness as an acceptance criteria, and many offer incentives to contractors for building smoother pavements. Ride quality is generally specified and tested with one of the following four methods: Straightedge measurements, Profilograph, Response Type Road Roughness Measuring Systems (RTRRMS), or Inertial profilometers. All four methods have been used by agencies for acceptance testing of roadway smoothness. Straightedge and Profilograph measurements have also been used for quality-control testing. Each of the four pavement smoothness measuring methods are discussed briefly below.

Straightedge

Straightedge measurements are perhaps the oldest form of measuring pavement smoothness, but are generally included in even the most modern specifications, and they may be required for both quality-control and acceptance testing. Quality-control checks of the pavement surface can be made with a straightedge during the compaction operation, providing immediate feedback to the paving-machine and compactor operators. Acceptance criteria using straightedge measurements generally include requirements for joints, pavement edges, and miscellaneous paved areas that are not accurately measured by other smoothness equipment.

Handheld straightedges range in length from 8 to 15 feet (2.4 to 4.6 meters), and rolling straightedges can be 30 feet (9.2 meters) in length or more.

Profilograph

Profilographs measure the longitudinal profile of pavements. They are used extensively for both quality-control and acceptance testing of HMA pavements. Profilographs measure smoothness with a profile index that is reported in inches per mile. In a rough sense, the inches per mile are the number of inches of bumps or depressions that are accumulated in a mile. The units are generally hand operated, and can be used in short lengths. They can be used for quality-control or acceptance, but not for network level pavement management.

Two types of profilometers are commonly used. They are referred to as the "California type profilograph" and the "Rainhart type profilograph." The two types of equipment do not give the same results, so it is important to use the type that is specified for the project.

Response Type Road Roughness Measuring System

Response Type Road Roughness Measuring Systems (RTRRMS) gauge smoothness by the up and down movement of a standard car in response to changes in road profile. This approach is intended to be what a traveler would feel if driving the roadway. As such it is susceptible to variations based on items such as speed, acceleration, and vehicle suspension. Readings are taken at standard speeds to simulate actual driving conditions.

The Mays Meter is the most common form of RTRRMS. The units of roughness are inches per mile. This equipment would not give the same values of inches per mile as the profilographs. Mays Meter values are used by agencies for acceptance and in pavement management systems to measure changes in pavement condition over time.

Inertial Profilometers

Inertial profilometers measure the actual profile in a longitudinal direction with readings typically taken along every inch of the roadway. They measure profile changes with laser or acoustic devices that are typically mounted on vehicle bumpers. They utilize inertial systems (accelerometers) that remove the up and down movement of the vehicle suspension from the data. Testing can be done at highway speeds (between 10 and 70 mph (6.2 kph and 43.4 kph) and the roadway roughness is presented in

units of International Roughness Index. There are several manufacturers of inertial profilometers, and AASHTO has developed provisional standards for calibrating and standardizing the equipment. The procedures are intended to enhance the accuracy and repeatability of tested values.

Inertial profilometers can be sensitive to the location and the number of the measuring sensors on the test vehicle. This is because the roadway profile may be different in the wheelpath, for example, than it is in the center of the lane. One way to increase repeatability is to add sensors and use an average profile. Repeatability between multiple profilometers generally requires the profilometers to have multiple sensors. With enough sensors, and careful driving path selection, roadway rutting can be measured by the profilometer.

Inertial profilometers are used extensively by agencies for acceptance testing of pavement smoothness and for pavement management data collection. Historically, they have not been used for quality-control purposes, but manufacturers have now developed lightweight devices mounted on small four-wheeled vehicles that can be used almost immediately after construction at slower speeds and for shorter distances.

10.8 Acceptance Criteria

The criteria for acceptance of HMA construction projects are established in the project specifications. Various types of specifications are used—and others are being proposed—that emphasize different levels of accountability and risk to construction contractors. The four specifications listed below can be viewed as a spectrum representing the progression of development for HMA specifications in the United States. They are the following:

- Method, or recipe, specifications
- End-result specifications
- Quality-assurance specifications
- Performance-type specifications

Method, or recipe, specifications detail the products and procedures used in the construction process. Equipment types and quantity, material criteria such as aggregate gradations, methods of construction such as number of roller passes, and other items closely control the construction process.

End-result specifications move away from controlling the process of construction and concentrate on the testing of the mixture and pavement, after completion, for specification conformance. The intent of this type of specification is that the quality criteria used to judge final acceptance be selected based on their impacts on maintenance costs and long-term pavement performance.

Quality-assurance specifications, while similar to end-result specifications, are based on statistical sampling and testing performed by the contractor and the purchasing agency. Quality-control procedures conducted by the contractor are used to assure that the specifications are being met throughout the process. Acceptance sampling and testing, performed by the agency, check the contractor's QC process and determine whether the materials should be accepted. The acceptance tests are based on appropriate quality criteria, as are end-result specifications. In addition, Independent-assurance (IA) sampling and testing are performed by a third party to validate the acceptance and/or quality-control results.

Performance-type specifications emphasize performance of the finished product over time and do not concentrate on construction activities or materials properties. Specific performance criteria are established in the specifications, which are based on what the finished roadway must meet after a specified length of time. Warranty specifications would fall in this category.

In addition, specifications that use performance-related tests to evaluate the ability of a mix to perform as intended over a period of time are considered to be performance-based specifications. Mathematical models are used to calculate the future performance of the pavement based on the test results from construction. Further discussion of performance specifications can be found in Section 10.9.

In evaluating acceptance test data, it is important to realize that the results of testing contain recognized statistical variation. In order to understand what the data means, the engineer cannot ignore the statistical nature of the results. The variations occur because of recognized limitations in the precision and the consistency of test equipment and procedures. Statistical acceptance methods are described in Section 10.8.4, "Statistical Acceptance," but even for traditional acceptance criteria, an understanding of the statistical nature of test results cannot be overlooked. It is necessary to have an understanding of the normal variation in specific test parameters to make reasonable decisions about acceptance and/or rejection of materials.

10.8.1 Historical Acceptance

Acceptance criteria are usually based on a number of specific properties of the mixture and the finished pavement. Historically, the following set of items was used to evaluate HMA construction projects:

- Materials—aggregate gradation, asphalt binder viscosity properties, and asphalt content
- Thickness tolerances—cores, yield checks, and string lines
- Pavement density—cores and nuclear gauge readings
- Pavement smoothness—straightedge

For many years, the responsibility for testing and controlling materials was held by the specifying agency. Contractors often submitted proposed aggregate and asphalt materials to the agency. The agency then performed a mix design and dictated the mixture proportions. At the start of production, the agency conducted asphalt content tests and aggregate gradations to make sure the job-mix formula was being duplicated within allowable tolerances. As long as the job-mix formula was adhered to, the assumption was made that the mixture volumetric criteria were also being met. This was often a mistaken assumption.

This type of combined process control and acceptance also required the agency to be responsible for adjusting the aggregate proportions and/or the asphalt content of the mixture. By changing the mixture proportions at the plant, the agency actions potentially had a direct effect on the contractor's ability to achieve density in the finished pavement. In some cases, penalties were assessed to the contractor for low densities even though the contractor was not in a position to make changes to the mixture to remedy the situation.

10.8.2 Modern Acceptance (with Volumetrics)

There have been recent trends toward quality-assurance specifications. Current trends also are emphasizing mixture volumetric properties over the

individual material components. Owners/agencies expect that including volumetric properties in acceptance criteria will lead to better performance of the pavement. As such, mix-design methods used today place more emphasis on volumetric properties such as air voids, VMA, and voids filled with asphalt.

Volumetric properties must be established in the mix design and controlled in the field. Nevertheless, gradation and asphalt-content determinations are also important to enable educated adjustments to the production process.

Process control has also been incorporated into the quality-assurance programs and the specifications of many agencies. The intent is to give the contractor more control over the finished product. Changes made at the asphalt plant, which affect the volumetric properties, placement, and compaction operations, are made by the contractor.

The quality of the finished pavement, however, is not entirely based on mixture properties. The density of the roadway after construction plays a major role in the overall performance of the roadway.

10.8.3 General Compaction Requirements

In most cases, pavement-density specifications are used to judge the acceptability of the compaction process during construction. The goal of compacting an HMA pavement is to achieve an optimum air-void content and provide a smooth, uniform surface. The resultant in-place air-void content of the HMA is a very important factor that affects performance of the mixture throughout the life of the pavement.

There are a variety of factors that are not part of the compaction operation that can have a significant impact on compaction. Many of these factors are listed below:

- Properly designed plant-produced mixtures
- Proper sampling and handling procedures of samples from the mixing facility or roadway
- Proper field laboratory testing procedures, especially correct compaction techniques and maximum theoretical density as appropriate
- Proper sampling, handling, and testing of the pavement core samples
- Adequate underlying support for the mixture during the compaction process

Most pavement compaction specifications fall into two categories. The first and most common specifies that compaction shall achieve a certain percentage of a reference density. This is called a reference density specification, and the reference density can come from the laboratory or from test strips in the field. The second category is called a method specification. This type requires that specified means and methods be accomplished. Each of these types of specifications will be discussed in more detail on the following pages.

Reference Density Specifications

Typical density specifications represent a comparison between the in-place density of the pavement that is achieved after final compaction, and a "reference" density. One of three reference densities is typically used in density specifications: maximum theoretical density, laboratory bulk density, or control-strip density. Use of the maximum theoretical density to determine HMA pavement density compliance is preferred by the Asphalt Institute.

The relationship between the reference-density measurements and the air voids of the in-place pavement is shown against each type of reference density. Eight percent is selected here because it is believed that if this level of compaction is achieved at the time of construction, 4 percent air voids will be achieved in a few years after further densification of the pavement under traffic (see Figure 10.14). Some agencies may prefer a target less than 8 percent air voids in the compacted pavement. A target higher than 8 percent is not recommended.

It should be noted that while the comparison between maximum theoretical density and in-place air voids is a consistent one, the relationships between the other two reference density types and in-place air voids will shift up or down depending on the actual mix design and compaction criteria used in the specification. For example, if the mix-design air void content is 5 percent as shown in Figure 10.15, then 100 percent of laboratory

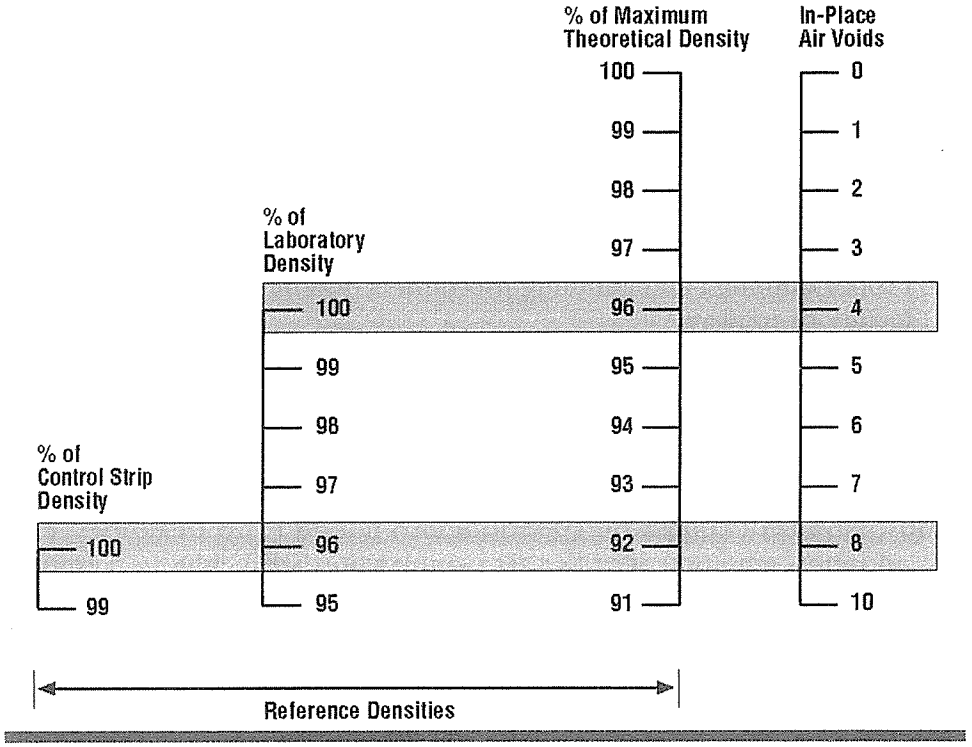


Figure 10.14 Reference Density at 4% Air Voids

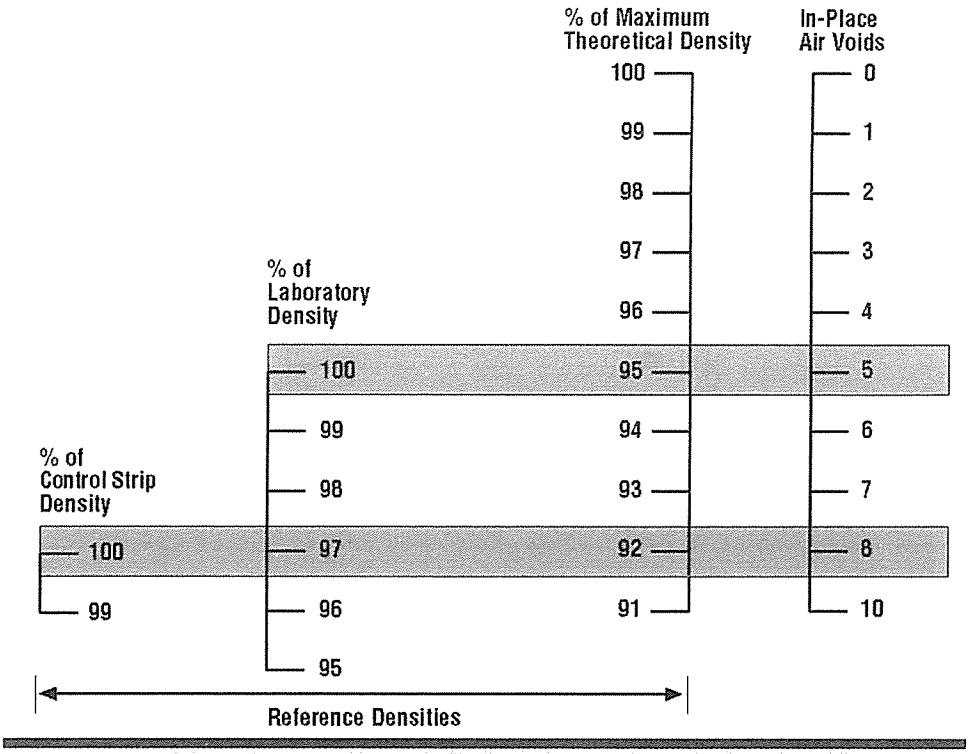


Figure 10.15 Reference Density at 5% Air Voids

compacted density would be at 5 percent air voids. The required percent laboratory density to achieve 8 percent in-place air voids is 97 percent. If a compaction criterion of 96 percent of laboratory density were used, an in-place air void content of 9 percent would result (not 8 percent).

Similarly, for a mix design air void content of 3 percent, as shown in Figure 10.16, a compaction criterion of 95 percent of laboratory density is required to achieve in-place voids of 8 percent. The compaction criterion of 96 percent would result in 7 percent in-place voids.

It is important to understand the relationships between the mix-design air void content and its effect on reference density specifications. A particular density specification may provide satisfactory field compaction on one project, but due to changes in mixture design criteria, may be inadequate on another.

Reference density specifications are generally used for pavements with a lift thickness greater than about 1 inch (25 millimeters), and some agencies use them for thin lift paving as well. The agencies' decision of which mix design air-voids and compaction limits are to be used should be based on many parameters including environmental conditions, traffic volume, subgrade support, size of the project, construction and testing schedules, and any lift thickness variation.

When a specific reference-density procedure (laboratory, maximum theoretical, or control strip) is chosen for a project, this same reference-density process should be used throughout the testing and determination of in-place

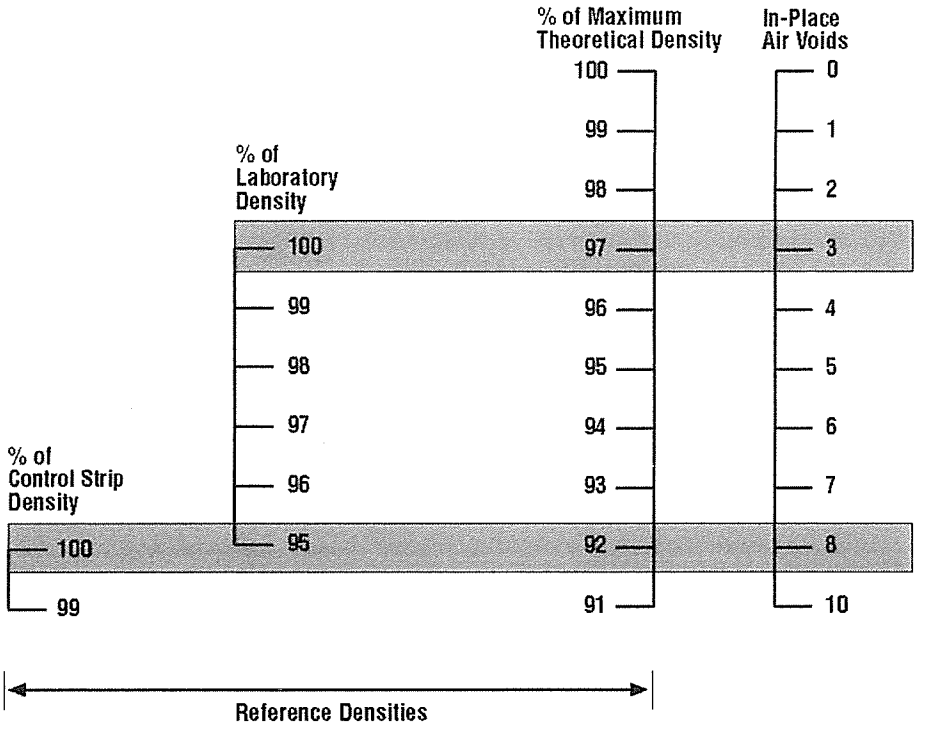


Figure 10.16 Reference Density at 3% Air Voids

density. This will ensure that a valid comparison exists in the determination of density compliance. A higher degree of compaction monitoring is necessary in the initial stages of the construction process, regardless of the density specification used, to ensure optimum results from the compaction process.

In addition to minimum compaction, it is also necessary to avoid too much compaction. Typical compaction operations will not compact an in-place mixture more than the laboratory compactor. If high density (less than 3 percent air voids) in a compacted mix is noted, the cause should be determined and corrected. This problem may also require that the mix be adjusted or redesigned. Pavement with an initial air void content below 3 percent is susceptible to permanent deformation and flushing after further consolidation due to traffic. Conversely, high initial air void levels, above 8 percent, are likely to yield pavements that age prematurely. These low-density pavements are more permeable to the damaging effects of air and water migrating into the interior of the mat. The result is a brittle pavement, which is more likely to crack due to age hardening of the asphalt binder and the effects of thermal expansion and contraction of the aggregate and moisture, resulting in a reduction in pavement strength.

For these reasons, it is recommended that compaction specifications require the resulting in-place air voids, immediately after construction, to be between 4 and 8 percent (96–92 percent of maximum theoretical density). Studies have shown this range in pavement air voids offers adequate resistance to air and water intrusion for most dense-graded

mixtures while allowing consolidation under traffic to take place without severe permanent deformation or flushing. Studies have also shown that extremely coarse-graded mixtures may need to be compacted to 6 percent air voids or less. The elevated coarse particle content in the absence of fine material allows the air voids to remain interconnected at a lower air void content requiring the increased compactive effort. In-place air voids for specialty mixtures such as stone matrix asphalt (SMA) or open-graded friction courses may require different limits.

Control Strip Density Specifications

Some agencies specify compaction requirements to meet a percentage of a control strip density. Under these specifications, a test strip of HMA is placed and compacted at the beginning of normal production paving. The control strip is placed using specified or otherwise selected rolling procedures and equipment. The intent is that the test strip be compacted to the maximum achievable density, which then becomes the specified reference density. Once the density of this control strip is established, the rest of the paving is required to achieve a percentage of this density. These types of specifications usually require the control strip to achieve a minimum percentage of the laboratory density as well, to avoid mixtures that are not compactable. This type of specification allows the relative difficulty (or ease) of compacting a given mix to become a factor in establishing the compaction requirements. New control strips are required for a different mix design or pavement application.

Quality-control testing for this type of project includes in-place density testing as described in 10.6.2, except that the compaction is calculated differently.

As an example, consider a project that specifies compaction be at least 98 percent of control strip density, and the control strip density measured for this particular application is 140.6 lb/ft^3 (2251 kg/m^3). If the quality-control testing yields in-place density of 139.2 lb/ft^3 (2228 kg/m^3) at a given location, then the compaction is calculated as follows:

$$\text{Compaction} = \left[\left(\frac{139.2}{140.6}\right) \times 100\right] \text{ or } \left[\left(\frac{2228}{2251}\right) \times 100\right] = 99.0 \text{ percent}$$

This compaction test result would comply with the specified minimum value of 98 percent.

Method Density Specifications

Some agencies specify detailed means and methods to be followed for compacting HMA. These methods include requirements such as size and type of compaction equipment, and they usually include a specific number of passes with each roller. Such specifications are useful when placing materials for which accurate density measurements are not easily made. Examples of this are leveling courses and small quantity placements in areas that carry little or no traffic.

Quality control for projects using method density specifications might include observing and documenting compliance with the specific

project requirements. Daily documentation, for example, of the compaction equipment utilized on the grade along with observed counts of the number of passes and temperature measurements may prove useful in verifying and documenting that the correct procedures were accomplished.

10.8.4 Statistical Acceptance—Percent Within Limits (PWL) Process

Acceptance test results on many projects are evaluated using statistical acceptance procedures. Pay factors may be calculated on specified criteria to award incentives or disincentives on the criteria that are evaluated. Sometimes pay factors may be calculated with a statistical process called Percent Within Limits (PWL), which is calculated for a given lot of material produced. The materials produced and placed for each lot would have pay factors that would be applied to the quantities produced in that lot.

Pavement owners wish to get their most important quality parameters as close as possible to the target values. The asphalt content of a mixture, for example, will be detrimental to the longevity of a pavement if it is either too high or too low. So the owners want plants producing HMA for their projects to be focused on achieving the actual target value of the job-mix formula, as opposed to achieving some minimum value. Statistically speaking, this means the owner wants the sample mean to match the target values. The sample mean is the average value of all test results. If four asphalt content tests were run for a given lot, for example, the average of those four tests would be the mean asphalt content for that lot.

The owners also want consistency of the materials produced. They want all the material produced to be close enough to the target values so that the long-term performance of the entire pavement will be good. Roadways often have to be replaced when the worst portion of the material begins to fail. Statistically speaking, the owner wants to have low variability or low variance in the test results. Variance is the difference between a tested value and the average tested value. So, specifying low variance is the same thing as saying all of the test results should be close to the average. Statisticians calculate a "standard deviation" to represent the variability in test results.

The third thing that owners want is to have a higher level of confidence that they know the level of quality actually achieved on the project. The test results from one compaction test for 5,000 feet of roadway, for example, would not give an engineer a high degree of confidence that he knows what compaction level was achieved throughout the pavement. A higher degree of confidence would be achieved if the appropriate number of samples were taken and the best (most accurate) sampling and testing methods available were used.

Statistical Basis

It is important to realize that the test results do not always represent the exact values that exist within the mixture. Assume, for example,

that an asphalt plant produces hot mix asphalt during a given day and the binder content starts out at 5.1 percent, increasing to 5.5 percent at the end of the day, but the overall average of the binder content is exactly 5.3 percent. Assume the laboratory gets two samples from the plant, one in the morning, and one at midday, and their test results are 5.1 percent and 5.3 percent. In this example, the average of the test results is (5.1 + 5.3)/2 = 5.2. The tested average was 5.2 percent and the actual average was 5.3 percent. This example also shows that if the laboratory had taken more samples, and had the samples been spaced more evenly to better represent the day's production, it could have come closer to achieving the actual average value.

This concept, that the test results do not exactly represent all of the material produced, is important for understanding the statistics. It is the reason that a statistical parameter called percent within limits (PWL) is calculated. If the asphalt content is measured on four samples evenly spaced throughout the day and all four of the tested asphalt contents fall within the upper and lower limits of the specification, intuitively that means we had 100 percent within the limits. Statistically speaking, however, that means there were four test values that can describe something about the mean asphalt content and the standard deviation of the asphalt content. Based on those numbers (mean and standard deviation), the percentage that fell within the upper and lower limits can be calculated. That percentage is the PWL, and it is calculated for each set of test results to determine what percentage, of all of the mix produced, likely fell within the specifications limits. The more test results there are, the more accurately the PWL represents the materials produced. If an insufficient number of samples are tested, the PWL will be inaccurate. If an agency desires to reduce the number of samples taken, the number of samples per lot should not be reduced. Increasing the lot size would be more appropriate.

Statistical acceptance procedures assume reasonably uniform production. The statistical models will not adequately represent materials' properties if a significant change occurs in the production process. This can be remedied by varying a lot size or truncating a lot when a known change is made to the production process.

Quality Characteristics for Pay Factors

Projects that are poorly controlled will have high variability in test results. The more consistent the materials and methods are, the lower the variability will be. Specifications establish acceptance parameters that will be used in pay-factor calculations. Attributes often used for acceptance include asphalt content, aggregate gradation, mixture volumetric properties, compaction, and ride quality. Agencies do not use statistical methods for all their acceptance criteria. Only parameters that can be tested and produced within reasonable statistical variation can be used. Segregation, for example, is not a parameter that can be effectively incorporated into a statistics-based acceptance program. Test procedures that generate repeatable numbers are needed for application of statistical methods.

The PWL is calculated using the sample mean, the sample variation, and the number of samples tested. The bottom line is that if contractors and producers can get the average values close to the target and produce and place mixtures with good consistency (low variation), then the result will be a high percentage of material within the specification limits (high PWL).

Some agencies use percent defects (PD) instead of PWL, but the statistical concepts are essentially the same. The PD is generally 100-PWL.

Specification Limits on Quality Characteristics

Specifications establish upper and lower limits for each of the quality characteristics that will be measured. These are defined as the upper specification limit (USL) and the lower specification limit (LSL). For example, if the asphalt content specifications set limits at 5.1 percent to 5.9 percent, then 5.1 percent is the lower specification limit and 5.9 percent is the upper specification limit.

The state of Florida studied testing and material variability on their HMA projects and established upper and lower specification limits for a variety of test parameters. Table 10.8 presents a simplified example of specification limits applied to different quality characteristics based on the general findings of the Florida study.

The examples of limits presented in Table 10.8 were derived from published limits from the Florida Department of Transportation based on variability of their statistically controlled production after it had been implemented for several years. Each agency or owner should establish their own limits based on their own expected variability of materials. Projects with low production, reduced QA/QC efforts, or with contractors and agencies inexperienced in PWL projects may necessitate wider limits.

Calculation of Percent Within Limits

Statistical acceptance methods use four numbers in evaluating the variations of each set of test results are acceptable. Those statistical numbers are number of test values (n), mean (x), standard deviation (s), and the percentage within the specification limits (PWL). Most of the tested properties of HMA will form a normal distribution curve when graphed on

Table 10.8
Example Specification Limits on Quality Characteristics

Quality Characteristic	Target Value	Lower Specification Limit	Upper Specification Limit
AC Content (%)	JMF	JMF−0.4	JMF+0.4
Passing #8 (%)	JMF	JMF−3.1	JMF+3.1
Passing #200 (%)	JMF	JMF−1.0	JMF+1.0
Air Voids (%)	4.0	2.6	5.4
Compaction, % G_{mm}	94.5	93.2	95.8

a histogram plot. This allows the use of mathematical formulas and charts that have been derived from the normal distribution function. Calculation of percent within limits for a given lot requires the following input values for calculation:

- Individual test values. (If there were four tests, there would be four results, identified as X1, X2, X3, and X4.)
- Number of test results (n), for four test results $n=4$
- Lower specification limit (LSL)
- Upper specification limit (USL)

From these values, the following formulas are used to calculate the intermediate statistical parameters and finally to calculate the quality index factors (Q_L and Q_U) for that lot:

$$\bar{x} = \frac{\sum x_i}{n}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$Q_U = \frac{USL - \bar{x}}{s}$$

$$Q_L = \frac{\bar{x} - LSL}{s}$$

where

x_i = Individual test value for a given sample
$\bar{x}$ = Mean of the test values for the lot
S = Standard deviation of test values for the lot
Q_U = Quality index relative to the upper specification limit
Q_L = Quality index relative to the lower specification limit

The quality index numbers (Q_L and Q_U) and the number of test results (n) are then applied to statistical tables to determine values of PWL (see table 10.9).

- PWL_U is the percent within the upper specification unit (based on Q_U).
- PWL_L is the percent within the lower specification unit (based on Q_L).
- PWL_T is the percent within the upper and lower specification limits.

$$PWL_T = PWL_U + PWL_L - 100$$

The total percent within limits (PWL_T) is necessary if there are both upper and lower limits. If there is only one limit, for example, a minimum compaction value was specified, then only one percent within limits would need to be determined. In the minimum compaction example, the percent within the lower specification limit (PWL_L) is used.

Pay Factors

Pay factors are then applied to the final values of the quality characteristic measured (usually PWL). Project specifications establish the pay factors and how they are to be applied. Some specifications use charts and others

Table 10.9
PWL Table (FHWA Format)

PWL	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$	$n = 9$	$n = 10$ to 11
100	1.16	1.50	1.79	2.03	2.23	2.39	2.53	2.65
99	–	1.47	1.67	1.80	1.89	1.95	2.00	2.04
98	1.15	1.44	1.60	1.70	1.76	1.81	1.84	1.86
97	–	1.41	1.54	1.62	1.67	1.70	1.72	1.74
96	1.14	1.38	1.49	1.55	1.59	1.61	1.63	1.65
95	–	1.35	1.44	1.49	1.52	1.54	1.55	1.56
94	1.13	1.32	1.39	1.43	1.46	1.47	1.48	1.49
93	–	1.29	1.35	1.38	1.40	1.41	1.42	1.43
92	1.12	1.26	1.31	1.33	1.35	1.36	1.36	1.37
91	1.11	1.23	1.27	1.29	1.30	1.30	1.31	1.31
90	1.10	1.20	1.23	1.24	1.25	1.25	1.26	1.26
89	1.09	1.17	1.19	1.20	1.20	1.21	1.21	1.21
88	1.07	1.14	1.15	1.16	1.16	1.16	1.16	1.17
87	1.06	1.11	1.12	1.12	1.12	1.12	1.12	1.12
86	1.04	1.08	1.08	1.08	1.08	1.08	1.08	1.08
85	1.03	1.05	1.05	1.04	1.04	1.04	1.04	1.04
84	1.01	1.02	1.01	1.01	1.00	1.00	1.00	1.00
83	1.00	0.99	0.98	0.97	0.97	0.96	0.96	0.96
82	0.97	0.96	0.95	0.94	0.93	0.93	0.93	0.92
81	0.96	0.93	0.91	0.90	0.90	0.89	0.89	0.89
80	0.93	0.90	0.88	0.87	0.86	0.86	0.86	0.85
79	0.91	0.87	0.85	0.84	0.83	0.82	0.82	0.82
78	0.89	0.84	0.82	0.80	0.80	0.79	0.79	0.79
77	0.87	0.81	0.78	0.77	0.76	0.76	0.76	0.75
76	0.84	0.78	0.75	0.74	0.73	0.73	0.72	0.72
75	0.82	0.75	0.72	0.71	0.70	0.70	0.69	0.69
74	0.79	0.72	0.69	0.68	0.67	0.66	0.66	0.66
73	0.76	0.69	0.66	0.65	0.64	0.63	0.63	0.63
72	0.74	0.66	0.63	0.62	0.61	0.60	0.60	0.60
71	0.71	0.63	0.60	0.59	0.58	0.57	0.57	0.57
70	0.68	0.60	0.57	0.56	0.55	0.55	0.54	0.54
69	0.65	0.57	0.54	0.53	0.52	0.52	0.51	0.51
68	0.62	0.54	0.51	0.50	0.49	0.49	0.48	0.48
67	0.59	0.51	0.47	0.47	0.46	0.46	0.46	0.45
66	0.56	0.48	0.45	0.44	0.44	0.43	0.43	0.43
65	0.52	0.45	0.43	0.41	0.41	0.40	0.40	0.40
64	0.49	0.42	0.40	0.39	0.38	0.38	0.37	0.37
63	0.46	0.39	0.37	0.36	0.35	0.35	0.35	0.34
62	0.43	0.36	0.34	0.33	0.32	0.32	0.32	0.32
61	0.39	0.33	0.31	0.30	0.30	0.29	0.29	0.29
60	0.36	0.30	0.28	0.27	0.27	0.27	0.26	0.26
59	0.32	0.27	0.25	0.25	0.24	0.24	0.24	0.24
58	0.29	0.24	0.23	0.22	0.21	0.21	0.21	0.21
57	0.25	0.21	0.20	0.19	0.19	0.19	0.18	0.18
56	0.22	0.18	0.17	0.16	0.16	0.16	0.16	0.16
55	0.18	0.15	0.14	0.14	0.13	0.13	0.13	0.13
54	0.14	0.12	0.11	0.11	0.11	0.11	0.10	0.10
53	0.11	0.09	0.08	0.08	0.08	0.08	0.08	0.08
52	0.07	0.06	0.06	0.05	0.05	0.05	0.05	0.05
51	0.04	0.03	0.03	0.03	0.03	0.03	0.03	0.03
50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Source: Specification Conformity Analysis, *FHWA Technical Advisory T5080.12*, June 23, 1989

use simple formulas. The AASHTO Quality Assurance Guide Specification presents the following equation to calculate the pay factor:

$$\text{Pay factor (\%)} = 55 + 0.5 \times \text{PWL}$$

This pay factor would then be multiplied times the contractor's payment for that particular bid item. Assume that the project specifications establishes that payment for HMA is by the ton completed and in place. For all of the tonnage represented by that lot, the price per ton would be paid at the bid price times the pay factor. This may cause the payment to increase or decrease depending on the contractor's accomplishment of the quality characteristics that were defined. Some agencies pay bonuses if the pay factor exceeds 100 percent. Other agencies restrict the total payment to 100 percent of the bid price. When such restrictions are in place, specifications can be written to allow positive pay factors to offset negative pay factors from previous paving lots.

In the pay factor equation above, the number "55" could be changed by the specifying agency. A closer look at the equation reveals that, if 55 were used, the highest pay factor that could be achieved would be 105 percent. This means a contractor could receive a 5 percent bonus if every lot achieved 100 percent within limits (PWL). Of course, if a PWL less than 90 percent were achieved on any given lot, the pay factor would be less than 100 percent, which would reduce the contractor's payment for that lot.

One can see that the agency can control the possible bonus by changing this pay factor variable. If the variable were set at 60, for example, the possible bonus would be 10 percent. A variable of 50 would mean that nothing greater than 100 percent pay could be achieved, but payment reductions would be assessed for any PWL less than 100 percent. Slight changes to this pay factor variable have severe financial impacts on the incentive or disincentive being offered to prospective bidders. The agency goal is to select a pay factor variable that will motivate contractors to achieve higher quality standards represented by high PWLs. In the end, this will cause the agency to pay a little more for a roadway that will last a lot longer.

Many specifications consider the fact that when mixtures are out of specification tolerances, they usually incur penalties based on more than one of the test requirements. Multiple non-compliances in one lot might unreasonably compound the penalties, or equal monetary consideration might be given to test parameters that do not hold equal value in performance. The percent passing the No. 8 (2.36 millimeters) sieve, for example, is not as important to pavement performance as air voids are. Some agencies are now applying weighting factors that consider all the statistical acceptance parameters for a given pay item, or identifiable group of pay items, when applying pay factors. Table 10.10 is an example of additional weighting factors that could be applied to pay factors for the production of hot mix asphalt. The pay factor distribution percentages shown in Table 10.10 could be used to calculate an overall pay factor by using the following equation:

$$\text{Pay Factor} = 55 + 0.5(.35(\text{PWL}_{AC}) + 0.10(\text{PWL}_{\#8}) + 0.20(\text{PWL}_{200}) + 0.35(\text{PWL}_{VOIDS})$$

Table 10.10
Pay Reduction Factors for HMA Production

Material Property	Pay Factor Distribution, percent
Asphalt Content, percent	35
Passing No. 8 (2.36 mm), percent	10
Passing No. 200 (0.075 mm), percent	20
Laboratory Voids, percent	35
Total	100

10.9 Performance-Based Specifications

Public agencies and other owners of asphalt pavements generally desire to construct smooth, long-lasting pavements that are cost effective when considering maintenance and all pavement costs. These needs have driven many innovative developments and are leading to different ways of purchasing HMA products.

Traditionally, HMA specifications have required measurement of basic quality characteristics that are believed to relate to long-term pavement performance. Pavement quality is defined by how well the measured properties of the constructed pavement match contract requirements. The specifications are focused on compliance with empirically derived quality parameters.

New "performance-based" specifications are being used that define quality based on predicted performance of the completed pavement. Payment is determined by comparing the life-cycle costs of the constructed pavement with those used in the pavement design. Two types of models are needed to calculate the life-cycle costs. The first are performance prediction models that estimate the severity, extent, and timing of distresses that will occur in the pavement. The second are maintenance cost models. They estimate life-cycle costs, including those for maintenance, repairs, and rehabilitation based on the distresses predicted.

10.9.1 Specification Definitions

Agencies and other owners of HMA pavements have used various types of specifications throughout the years. Specifications that use performance measures can be explained in the context of some of the other specifications that have been used. The first four definitions are from the AASHTO Highway Subcommittee on Construction, Quality Construction Task Force.

Method specifications are specifications that require the contractor to produce and place a product using specified materials in definite proportions and specific types of equipment and methods under the direction of the agency.

End-result specifications are specifications that require the contractor to take the entire responsibility for producing and placing a product. The agency's responsibility is to either accept or reject the final product or to apply a price adjustment commensurate with the degree of compliance with the specifications.

Performance-related specifications are specifications that use quantified quality characteristics and life-cycle cost relationships that are correlated to product performance.

Performance-based specifications are quality-assurance specifications that describe the desired levels of fundamental engineering properties (resilient modulus, creep properties, and fatigue) that are predictors of performance and appear in primary prediction relationships (models that can be used to predict stress, distress, or performance from combinations of predictors that represent traffic, environment, supporting materials, and structural conditions).

Performance Specifications are specifications that require the contractor to take responsibility for the actual performance of the product produced. These specifications define the time frame and establish clear performance standards. Performance specifications are also called Warranty Specifications and are addressed in Section 10.10.

10.9.2 Performance-Related Specifications

Performance-related specifications use quality characteristics already being measured and specified by many agencies. These characteristics are then correlated to the life-cycle cost of the pavement.

Three key points distinguish performance-related specifications from others. One is that the quality characteristics that are tested and used for acceptance are related to fundamental engineering properties of the pavement. Another is that the fundamental engineering properties use mathematical models to predict the extent and the timing of distress in the pavement. The third point is that the specifications use the distress prediction in mathematically predicted life-cycle costs for establishing contract requirements, including penalties and rewards.

The following quality characteristics might be seen in a performance-related specification.

- Segregation
- Ride quality
- Pavement thickness
- Asphalt content/effective asphalt content
- Pavement density (in place)
- Aggregate gradation
- Asphalt binder viscosity
- Longitudinal joint air voids
- Permeability (in place)
- Low temperature tensile strength
- Fracture temperature

10.9.3 Performance-Based Specifications

Performance-based specifications might be considered the next phase of performance-related specifications. They will be implemented in the same way, except that performance-based specifications will use test

methods that directly measure key fundamental engineering properties of the pavement. These measured properties will be used to predict performance and then used in the life-cycle cost models.

Performance-based specifications have not been used yet for HMA pavements. Models predicting performance have been developed and are currently being evaluated and validated. Many state highway agencies have begun refining the calibration of the models to reflect their local construction materials, specifications, and techniques.

The prediction models will work in tandem with the pavement design because the design input parameters will be the same engineering properties that will be measured on the constructed pavement. If significant differences are found in properties of the constructed pavement versus the design, the models could be rerun and effects on the life-cycle costs could be evaluated.

10.10 Performance Specifications—Warranties

Warranties are not new to construction contracts. In recent years, however, there has been a growing interest by many owners of asphalt pavements to get long-term warranties for road construction. Warranties, or performance specifications, require a contractor to warrant that an item will perform in a manner that has been agreed to in advance.

Successful performance specifications require the contractor to be responsible for those specific pavement characteristics that the contractor can control. Specifications that require contractors to warrant everything, including things well outside of the contractor's control or ability to estimate, are both unfair and unwise. For example, contractors generally have no control over traffic-loading, future development, etc. It is in the owner's interest to have clear definitions of the contractor's responsibilities under the warranty. The warranties are normally secured by bonds. For long warranty periods, the bonds can be very expensive, and become a significant part of the cost of the project. Because of this, warranty paving projects in the United States typically have been limited to about 5 years. Requiring a 20-year bond for full reconstruction costs may be unreasonable and/or very expensive.

Warranty contracts can require the contractor to provide project acceptance and quality-control criteria to the owner. Typically a quality-control plan including documentation of certifications for the materials used on the project will be included. The plan could provide certifications of the contractor's testing laboratory and personnel, mix design methods, sampling methods, plant operations, lay-down operations, and density control methods. Some warranty contracts also require a documentation process that will provide the agency with acceptance-type test reports and other pavement data that they would normally collect on their own projects, use with pavement management or other asset management databases.

10.10.1 Performance Characteristics

Clear identification of the required performance characteristics is a critical aspect of warranty specifications. Agencies often use key parameters tracked in their pavement management systems for acceptance criteria; therefore, smoothness, skid resistance, and pavement distress parameters are included. Performance criteria are typically established for:

- Rutting
- Fatigue cracking
- Other cracking
- Smoothness
- Skid resistance